

MOTA • ACADEMIC YEAR 2025-26

Opening the Gate

*An Impact Topography of Teacher Agency,
Mathematical Translation, and Community Multiplication*

A sustained Olympiad-to-Classroom community in which teachers move from learning advanced mathematics to translating it, teaching it, building with it, and reporting early learner-facing signals.

17	51	354	681
PARTICIPANT-FACING SESSIONS	SCHEDULED LIVE HOURS	RECORDED ATTENDANCES	PARTICIPANT-HOURS

Programme development was funded by SITE when it was part of the NYU Abu Dhabi Research Institute, with MIMS as the academic partner.

REPORT MAP

How to read this report

This report combines administrative programme records with a structured qualitative analysis of anonymized participant reflections. It separates what is directly established by records from what participants report, and it treats learner-facing statements as signals unless they are independently documented.

1	Programme model	Why sustained cycles and teach-backs place teacher agency at the centre.
2	Delivery and engagement	The verified AY 2025-26 cycle and attendance record.
3	Evidence architecture	Sources, case construction, coding levels, and analytic rules.
4	Impact topography	How to read evidence position and professional-action constellations.
5	Cross-case findings	Patterns across mathematical learning, agency, transfer, and learner-facing reports.
6	Participant evidence atlas	Sixteen anonymous case records with evidence trails and boundaries.
7	Evidence statement	What the current evidence establishes and what it does not establish.
A	Method appendix	Coding decisions, evidence status, and data dictionary.

EVIDENCE POSITION

The analysis does not use evidence levels as a prestige ranking. A higher number means that a statement reaches farther along the pathway from appreciation to learner-facing evidence. It does not mean that the claim is more important, independently verified, representative, or caused by MOTA.

EXECUTIVE SYNTHESIS

The evidence position in one page

MOTA's AY 2025-26 evidence base shows a sustained professional-development architecture operating beyond delivery alone. Four cycles combined 13 coach-led sessions with four participant presentation sessions. Across 17 participant-facing sessions, programme records document 51 scheduled live hours, 354 recorded attendances, and 681 participant-hours.

The reflection dataset contains 17 submissions representing 16 unique participants after two submissions from one participant were consolidated. Fifteen of the 16 cases describe a specific classroom, training, community, competition, or system use. Twelve contain at least one learner-facing signal, such as reported persistence, strategy selection, mathematical explanation, student production, participation, or competition performance. These are case-level reports; they do not establish representative or causal downstream outcomes.

DELIVERY	17 participant-facing sessions · 51 scheduled hours	ESTABLISHED
ENGAGEMENT	354 recorded attendances · 681 participant-hours	ESTABLISHED
TEACHER AGENCY	Teach-backs plus reports of explaining, adapting, designing, training, and coordinating	SUPPORTED
TRANSFER	15 of 16 cases contain specific enacted classroom, training, community, or system use	SELF-REPORTED EVIDENCE
DOWNSTREAM OUTCOMES	12 of 16 cases contain learner-facing signals	NOT ESTABLISHED

The central pattern is translation. Participants describe selecting methods, simplifying content, redesigning tasks, preparing teach-backs, training others, creating or coordinating competitions, and moving ideas into national-team and community settings. The strongest evidence is specific about the mathematical idea and the professional action. The most important boundary is equally specific: much of the learner-facing evidence is self-reported, and referenced student work, assessments, competition files, results, and reach records were not supplied for independent inspection.

CURRENT CONCLUSION

Delivery and engagement are established by programme records. Teacher agency and enacted transfer are supported by the Academy's teach-back structure and by specific participant reports. Learner-facing signals are present in 12 cases; independent downstream outcomes are not established.

1 • PROGRAMME MODEL

Opening Olympiad mathematics through teachers

Olympiad mathematics is often organized as a selective culture: difficult problems identify a small group of highly prepared students, while the structures behind those problems remain distant from ordinary classroom experience. MOTA starts from a different proposition. Invariants, extremal reasoning, combinatorial structure, strategic experimentation, proof, and advanced geometry can become sources of investigation for wider groups of learners when teachers have the mathematical range and professional agency to translate them.

The Academy therefore works through teachers. Deeper content knowledge expands the explanations, representations, questions, and pathways a teacher can offer. The programme then adds a second requirement: participants are not positioned only as recipients of expert mathematics. They are expected to interpret, present, adapt, and carry mathematical ideas into their own settings.

The sustained-cycle architecture

AY 2025-26 was organized as four deep-dive cycles running approximately biweekly. Sessions were scheduled for about three hours. Each cycle combined several coach-led sessions with a participant presentation session. The coach developed a mathematical domain; participants then returned to explain what they had learned and how the material might become an activity, lesson, training sequence, problem set, assessment, or community initiative.

THE STRUCTURAL MOVE

The teach-back is part of the learning design. Preparing to explain an idea requires a participant to select what matters, reconstruct the reasoning, anticipate an audience, and take responsibility for how the mathematics will travel.

A theory of action grounded in translation

- Expert mathematical study expands the participant's range of methods and representations.
- Repeated engagement and teach-backs rehearse interpretation, explanation, and professional authorship.
- Participants adapt mathematical structures for classrooms, camps, training programmes, competitions, and professional communities.
- Those uses may generate learner-facing changes in strategy, persistence, explanation, participation, or production.
- Independent downstream outcomes require evidence beyond participant reflection, such as supplied artifacts, student work, assessments, observations, or verified administrative records.

This theory of action explains why the report treats teacher agency as a mechanism rather than a slogan. Agency becomes visible when a participant chooses, reconstructs, designs, coordinates, trains, or creates—not merely when a participant reports satisfaction.

2 • DELIVERY AND ENGAGEMENT

A complete academic-year cycle record

The verified AY 2025-26 schedule contains 13 coach-led sessions and four participant presentation sessions. Together they form 17 participant-facing sessions across four cycles.

Cycle	Coach-led sessions	Participant presentations	Participant-facing total
Cycle I	4	1	5
Cycle II	3	1	4
Cycle III	3	1	4
Cycle IV	3	1	4
Total	13	4	17

Recorded engagement

17	51	354	681
PARTICIPANT-FACING SESSIONS	SCHEDULED LIVE HOURS	RECORDED ATTENDANCES	PARTICIPANT-HOURS

An attendance is one recorded participant presence at one session. Participant-hours aggregate the recorded time contributed by participants across sessions. These measures establish delivery volume and engagement; they do not by themselves show professional change or student impact.

What the numbers make visible

The programme's distinctive feature is the repeated time structure. The relevant unit is not a one-off webinar or conference encounter but a sequence in which participants return, study a domain over several sessions, and present. The 51 scheduled hours and 681 participant-hours make the duration of that structure visible. The reflection analysis then asks whether participants describe movement from attendance into agency and use.

ADMINISTRATIVE FACT

The figures on this page come from programme schedule and attendance records. They are not estimated from the reflection form.

3 • EVIDENCE ARCHITECTURE

From 17 submissions to 16 anonymized cases

The qualitative source contains 17 reflection submissions. Two submissions came from the same participant and were combined to preserve one analytic case per participant. The resulting dataset contains 16 cases, labeled P01-P16. Names and email addresses are not used in the analysis or in the case atlas.

Evidence sources

- Programme schedule and attendance records for the four AY 2025-26 cycles.
- Seventeen teacher-reflection form submissions consolidated into 16 unique cases.
- A small number of supplementary participant messages linked to the relevant anonymous case where the connection was identifiable.
- References within responses to lessons, competitions, assessments, projects, photographs, videos, question files, results, and reach figures. Unless supplied and inspected, these remain referenced rather than independently documented artifacts.

The six-level response framework

Level	What the response supports	Interpretation	Boundary
L0	Appreciation, satisfaction, or insufficient specificity	Testimonial only	Does not identify learning or change
L1	Claimed mathematical or professional learning	Self-reported learning	Does not show enacted practice
L2	Intention or preparation to change practice	Planned transfer	Does not show enactment
L3	Specific classroom, training, community, or system use	Self-reported enacted transfer	Use is not independently observed
L4	Concrete material or artifact associated with use	Documented transfer only when supplied and inspected	A reference to a missing artifact is not audited evidence
L5	Student or trainee behavior, work, participation, or outcome	Learner-facing signal	Self-report does not establish prevalence or causality

L4 STATUS IN THIS DATASET

One case contains statement-level references to a competition and an assessment. The materials themselves were not embedded in the analyzed source and were not inspected. They are therefore reported artifact references, not independently audited documentation.

3 • EVIDENCE ARCHITECTURE

How each case was read

1	Consolidate	Combine repeat submissions from the same participant before counting cases.
2	Unitize	Separate a response into meaning units when it contains more than one kind of evidence.
3	Code levels	Assign each meaning unit the lowest defensible level supported by its wording.
4	Record the highest position	Place the case at the farthest supported level while retaining all lower-level evidence in its trail.
5	Code professional action	Identify classroom translation, community multiplication, system building, or emerging evidence. Memberships may overlap.
6	Assess MOTA connection	Distinguish an explicit link to Academy content from mixed, weak, or absent linkage.
7	Record artifact status	Separate a described or referenced artifact from a file or record actually supplied for inspection.
8	Preserve attribution boundaries	Retain participants' own cautions and distinguish temporal association from causal attribution.

Two non-interchangeable conclusions

A case-level L5 code means that a participant described something learner-facing. A programme-level downstream-outcome conclusion asks a different question: whether learner outcomes have been established across an independently documented evidence base. The first is present in this dataset; the second is not.

DELIVERY	17 participant-facing sessions • 51 scheduled hours	ESTABLISHED
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4 • IMPACT TOPOGRAPHY

A map of evidence position and professional action

The topography combines two analytic structures. Distance from the centre shows the highest evidence level reached by a case. Colour shows the case's primary professional-action constellation. The map is descriptive: it does not rank teachers, measure effect size, or prove causality.

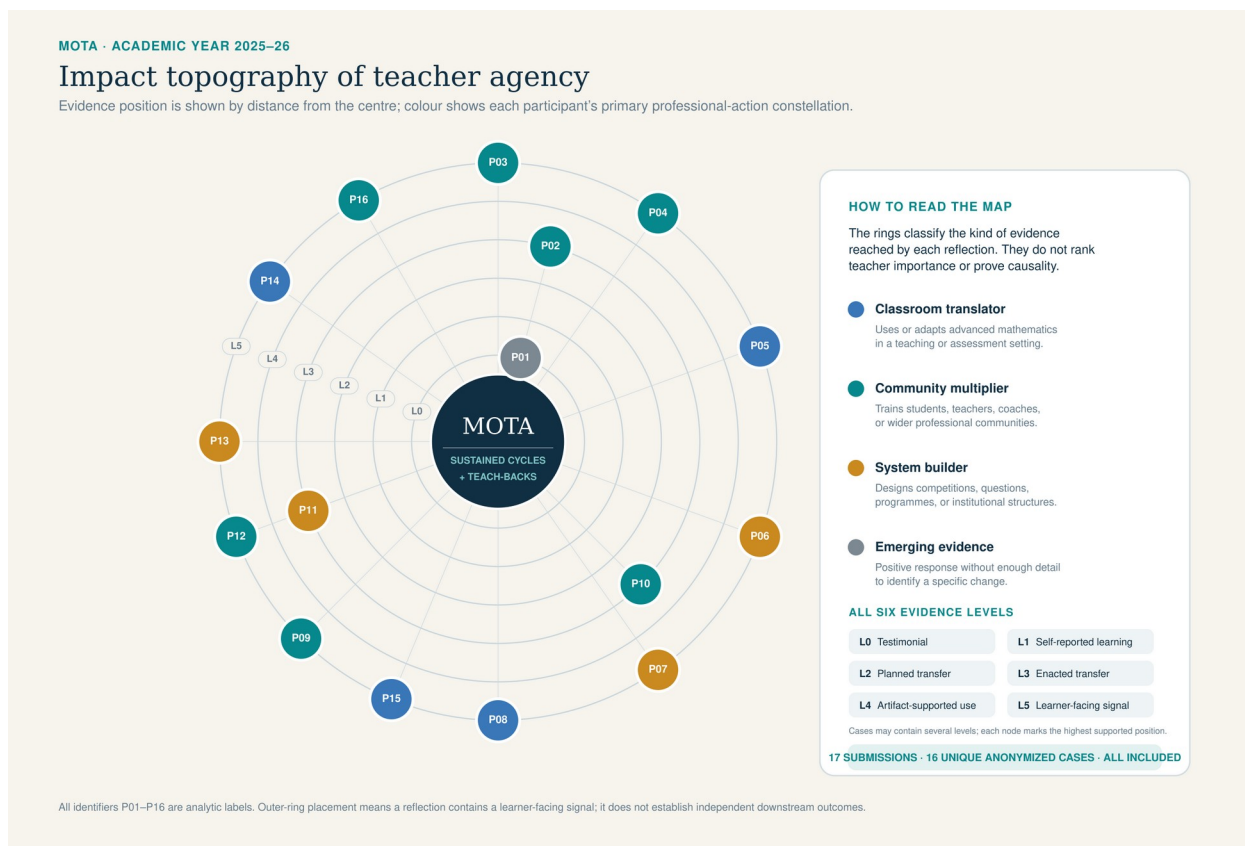


Figure 1. Sixteen anonymized participant cases. Outer-ring placement means that a reflection contains a learner-facing signal; it does not establish independent downstream outcomes.

4 • IMPACT TOPOGRAPHY

What the map contains

Level	Evidence position	Cases containing level	Cases ending at level
L0	Testimonial / insufficient specificity	2	1
L1	Self-reported learning	5	0
L2	Planned transfer	1	0
L3	Self-reported enacted transfer	15	3
L4	Artifact-referenced use	1	0
L5	Learner-facing signal	12	12

Cases may contain several levels, so the presence column is intentionally non-exclusive. The endpoint column counts each case once at its highest supported position: one case ends at L0, three at L3, and 12 at L5.

Professional-action constellations

Constellation	Primary	Any membership	What it captures
Classroom translation	4	12	Uses or adapts advanced mathematics in teaching, training, or assessment.
Community multiplication	7	11	Trains students, teachers, coaches, or wider professional communities.
System building	4	6	Designs or coordinates questions, competitions, programmes, or structures.
Emerging evidence	1	1	Positive response without enough detail to identify a change pathway.

Memberships overlap because a participant may translate mathematics for a classroom, train a wider community, and help build a competition or programme. Colour on the map shows only the primary constellation selected for visual clarity.

5 • CROSS-CASE FINDINGS

The recurring pattern is mathematical translation

The reflections do not merely list topics. The more detailed cases describe how a mathematical method entered a professional decision: what to emphasize, how to reduce difficulty, when to withhold a diagram, how to sequence a technique, what kind of hint to give, or how to turn content into a challenge, presentation, assessment, camp, or training programme.

Mathematical learning becomes method selection

Participants identify equality cases as a guide to AM-GM, the Cauchy-Bunyakovsky-Schwarz inequality as a restructuring tool, Fermat and Euler in last-digit work, cyclic quadrilaterals in geometric decomposition, and invariants or Nim in strategic investigation.

Agency appears as authorship and responsibility

Teach-backs, re-explanation, preparation of pedagogical content, adaptation of exercises, training of other coaches, and coordination of competitions show participants assuming responsibility for how mathematics is represented and transmitted.

Transfer is adaptation rather than replication

Participants report simplifying content for early-stage programmes, combining geometry with other representations, redesigning assessments, moving exercises into face-to-face and remote training, and using invariants to change the questions students ask.

Multiplication operates across professional settings

Cases describe classrooms, teacher development, camps, national-team training, cross-border coaching, question design, student projects, and competition structures. These settings are evidence of professional reach, not automatic evidence of MOTA causation.

Learner-facing signals are mainly behavioral

The most specific reports concern persistence, trying multiple strategies, asking where to look before calculating, explaining reasoning, seeking additional problems, applying named methods, producing projects, or discovering a winning strategy.

5 • CROSS-CASE FINDINGS

Where mathematical depth becomes usable

The dataset does not support a defensible ranking of cycles or topics. Respondents named domains in different ways, sometimes selected more than one cycle, and occasionally linked a topic to a cycle differently from the form labels. The analysis therefore describes recurring mathematical content without converting the reflections into a popularity table.

Combinatorial games and invariants

Participants describe searching for what remains constant, exploring winning strategies in Nim, and using games to shift attention from immediate calculation to structure.

Inequalities

Accounts mention AM-GM, equality cases, Cauchy-Bunyakovsky-Schwarz, Engel form, equivalent reformulation, and the selection of an appropriate inequality method.

Number theory

Fermat, Euler, congruences, primitive roots, LTE, Diophantine equations, and last-digit problems appear in learning, training, and challenge design.

Geometry

Cyclic quadrilaterals, incidence relations, power of a point, angle chasing, coordinates, barycentric methods, complex numbers, and regular-polygon construction appear across classroom and training accounts.

Problem-solving heuristics

Several cases emphasize orientation before execution: deciding where to look, comparing methods, delaying formula search, and sequencing hints to preserve learner ownership.

INTERPRETIVE BOUNDARY

Topic recurrence shows what participants chose to discuss. It does not measure topic quality, cycle effectiveness, or the proportion of all registered teachers who used each method.

5 • CROSS-CASE FINDINGS

Participant voice

The following excerpts are reproduced exactly from English-language form responses. Grammar and wording are unchanged. The labels are analytic identifiers only.

“One idea I now use is the invariant method from Cycle II. It changed the way I approach and teach problems by encouraging me to look for what remains unchanged rather than immediately searching for a formula. For example, when teaching problem-solving, I encourage students to identify patterns or properties that stay constant throughout a process, helping them develop deeper logical thinking and explore problems more creatively.”

P08 • LEARNING INTO CLASSROOM PRACTICE

“With this academy, we are not merely recipients. We must also be able to grasp, practice, and re-explain what we have learned from the experts. That is what is very important.”

P09 • TEACHER AGENCY

“Cycle IV - the Nim game - everything that i revised and give it to my students- while preparing the presentation”

P14 • TEACH-BACK INTO STUDENT USE

“I coordinated and prepared for the first Mathematics Olympiad in Anbar and was the supervisor of the competition and supervised the development of questions commensurate with the topics of the World Mathematics Olympiad.”

P11 • SYSTEM BUILDING

The quotations demonstrate specificity and professional voice. Their evidentiary status remains self-report unless the described use or learner response is supported by a supplied artifact, observation, assessment, or independent record.

6 • PARTICIPANT EVIDENCE ATLAS

Sixteen anonymized evidence trails

Each case page retains the evidence trail rather than reducing a participant to one score. The highest position determines map placement; lower-level learning, planning, and transfer statements remain visible. The MOTA-link field describes how explicitly the response connects the reported activity to Academy content or structure.

Link category	Meaning
Strong	The response explicitly links a method, exercise, presentation, or use to an Academy cycle or resource.
Mixed	The response contains an Academy connection alongside explicit uncertainty, concurrent activity, or a substantial pre-existing role.
Weak	The professional action is specific, but the response does not state that the Academy caused or materially shaped it.
None	The response is too limited to identify an Academy connection.

READING THE ATLAS

A detailed case can still have a cautious boundary. Specificity improves interpretability; it does not replace independent verification.

6 • PARTICIPANT EVIDENCE ATLAS | P01 • ANONYMIZED CASE

A positive but non-specific response

POSITION	L0 • Testimonial / insufficient specificity	MOTA LINK	None
PRIMARY	Emerging evidence	LEVELS	L0

The participant affirmed training, method use, a student moment, and an outcome, but supplied no example that could be interpreted without invention.

Professional context

Roles: Role not specified in the response

Mathematical domains: Number theory

Evidence trail

- L0 • Affirmative one-word responses indicate a positive experience but do not identify what changed.

Artifact status

None described.

INTERPRETIVE BOUNDARY

No learning, transfer, or learner outcome can be inferred from yes/no answers alone.

6 • PARTICIPANT EVIDENCE ATLAS | P02 • ANONYMIZED CASE

Operating across trainer development, coaching, and competition coordination

POSITION	L3 • Self-reported enacted transfer	MOTA LINK	Weak
PRIMARY	Community multiplication	LEVELS	L0, L2, L3

The participant described trainer development in Côte d'Ivoire, support for Tunisian and African students, work in Djibouti, coordination of PAMO 2026, and preparation of specialized Olympiad pedagogical content.

Professional context

Roles: Olympiad trainer; Competition coordinator

Mathematical domains: Geometry; Number theory; LTE

Evidence trail

- L3 • Delivered or supported trainer and student development across several African contexts and coordinated an international competition.
- L2 • Reported preparing specialized Olympiad pedagogical content for students.
- L0 • A supplementary message described the Academy as a high-quality opportunity for knowledge and skill development.

Artifact status

Specialized pedagogical content was described as in development; it was not supplied for inspection.

INTERPRETIVE BOUNDARY

The professional activity is specific, but the record does not establish that MOTA caused or materially changed it.

6 • PARTICIPANT EVIDENCE ATLAS | P03 • ANONYMIZED CASE

Using Academy algebra in training while resisting over-attribution

POSITION	L5 • Learner-facing signal	MOTA LINK	Mixed
PRIMARY	Community multiplication	LEVELS	L3, L5

The participant reported using Academy algebra material with students and intensifying remote contact. Students were described as using AI tools to investigate solutions and showing greater ambition to solve more problems and seek Olympiad resources.

Professional context

Roles: National-team trainer

Mathematical domains: Algebra; Geometry; Inequalities

Evidence trail

- L3 • Used material from the Academy's algebra work in student training.
- L3 • Applied remote communication with students more intensively.
- L5 • Reported AI-assisted solution searching, increased ambition, and greater use of Olympiad problem sites.

Artifact status

No student work or usage records were supplied.

INTERPRETIVE BOUNDARY

The participant explicitly cautioned that the student change may not be directly attributable to MOTA; that uncertainty remains part of the evidence.

6 • PARTICIPANT EVIDENCE ATLAS | P04 • ANONYMIZED CASE

Extending Academy geometry into multi-method teaching and student production

POSITION	L5 • Learner-facing signal	MOTA LINK	Strong
PRIMARY	Community multiplication	LEVELS	L3, L5

Two submissions describe a 20-day training programme, nine lectures, teacher development, and youth and national-team preparation. Academy geometry was combined with complex numbers, coordinate and barycentric methods, and sequences.

Professional context

Roles: Scientific-committee leader; Olympiad trainer; Academic supervisor

Mathematical domains: Geometry; Complex numbers; Sequences; Coordinate geometry; Barycentric methods

Evidence trail

- L3 • Reused Academy geometry and incidence methods within an extended sequence of lectures and training programmes.
- L3 • Withheld diagrams and digital tools so students had to construct representations and compare multiple solution routes.
- L5 • Students connected complex numbers and sequences, generalized regular-polygon constructions, and used induction.
- L5 • Twenty-five students working in five groups produced polygon, mosaic, and linear-programming projects.

Artifact status

Project files, photographs, and videos were offered but were not included in the source workbook and were not inspected.

INTERPRETIVE BOUNDARY

The account is unusually specific, but the described student products and media were not available for independent inspection.

6 • PARTICIPANT EVIDENCE ATLAS | P05 • ANONYMIZED CASE

Moving advanced heuristics into challenges and more creative assessments

POSITION	L5 • Learner-facing signal	MOTA LINK	Strong
PRIMARY	Classroom translation	LEVELS	L1, L3, L5

The participant reported introducing Academy-inspired challenges and making university examinations more creative. Students were described as becoming more interested and bringing Olympiad problems they had attempted.

Professional context

Roles: University mathematics professor

Mathematical domains: Inequalities; Number theory; Sequences; Geometry; Problem-solving heuristics

Evidence trail

- L3 • Introduced Academy-inspired problems and redesigned examinations to be more creative.
- L1 • Named specific heuristics for choosing inequality methods, constructing sequences, and angle chasing.
- L5 • Reported that students showed more interest and began bringing Olympiad problems they had tried to solve.

Artifact status

Creative examinations and Academy-inspired problems were referenced but not supplied.

INTERPRETIVE BOUNDARY

The response supports changed practice and student interest, not measured learning gains.

6 • PARTICIPANT EVIDENCE ATLAS | P06 • ANONYMIZED CASE

Transferring inequality methods into challenge design and trainer learning

POSITION	L5 • Learner-facing signal	MOTA LINK	Strong
PRIMARY	System building	LEVELS	L3, L5

The participant helped prepare Yemeni Olympiad challenges and trained Olympiad coaches. Cycle III and the Cauchy–Bunyakovsky–Schwarz inequality were identified as especially influential.

Professional context

Roles: Olympiad challenge designer; Coach trainer

Mathematical domains: Inequalities; Cauchy–Bunyakovsky–Schwarz inequality

Evidence trail

- L3 • Helped design national Olympiad challenges and trained other coaches.
- L3 • Reported using the Cauchy–Bunyakovsky–Schwarz approach to solve inequalities after Cycle III.
- L5 • A trainee transformed an inequality into an equivalent form and proved it through an unexpected valid approach.

Artifact status

The trainee's unexpected inequality solution was described but not supplied.

INTERPRETIVE BOUNDARY

The trainee's mathematical argument was not available for independent checking.

6 • PARTICIPANT EVIDENCE ATLAS | P07 • ANONYMIZED CASE

A direct content-to-system pathway across multiple Olympiad domains

POSITION	L5 • Learner-facing signal	MOTA LINK	Strong
PRIMARY	System building	LEVELS	L1, L3, L5

The reflection and supplementary message identify where Academy ideas entered two seasons of Yemeni Olympiad challenges, often at reduced difficulty for an early-stage programme.

Professional context

Roles: Olympiad challenge designer; Teacher educator

Mathematical domains: Inequalities; Combinatorics; Invariants; Nim; Geometry; Number theory

Evidence trail

- L1 • Explained equality cases as a guide to AM–GM and named CBS, Engel form, primitive roots, Fermat, and Euler as active tools.
- L3 • Mapped Academy topics to specific stages and question numbers across two Olympiad seasons.
- L3 • Reported simplifying Academy material for early seasons and sequencing advanced techniques gradually.
- L5 • Students independently discovered a first-player winning strategy for a two-pile Nim situation.
- L5 • The challenges were reported as shared with more than 200,000 students.

Artifact status

Two seasons of challenge files were referenced and said to be awaiting retrieval from an archive; they were not included in the analyzed source.

INTERPRETIVE BOUNDARY

The detailed mapping strengthens the MOTA connection. The challenge files and reach figure remain uninspected participant reports.

6 • PARTICIPANT EVIDENCE ATLAS | P08 • ANONYMIZED CASE

From asking for formulas to investigating what remains invariant

POSITION	L5 • Learner-facing signal	MOTA LINK	Strong
PRIMARY	Classroom translation	LEVELS	L3, L5

The participant used invariants to shift problem solving away from immediate formula seeking. A previously hesitant student tried several strategies, found a hidden pattern, and explained the reasoning to the class.

Professional context

Roles: Mathematics teacher

Mathematical domains: Combinatorial games; Invariants; Problem solving

Evidence trail

- L3 • Encouraged students to look for patterns or properties that remain constant instead of immediately searching for a formula.
- L5 • A previously hesitant student tried multiple strategies, identified a hidden pattern, and explained the reasoning to the class.
- L5 • The participant perceived improved final-exam grades and believed deeper problem solving contributed.

Artifact status

No student work or grade records were supplied.

INTERPRETIVE BOUNDARY

The student episode is a specific behavioral signal. The grade statement remains a perception because no comparison or grade data were supplied.

6 • PARTICIPANT EVIDENCE ATLAS | P09 • ANONYMIZED CASE

Professional agency as grasping, practising, and re-explaining

POSITION	L5 • Learner-facing signal	MOTA LINK	Strong
PRIMARY	Community multiplication	LEVELS	L1, L3, L5

An experienced Olympiad leader described MOTA as extending existing work through additional resources, expert learning, and an expectation that participants re-explain what they learn.

Professional context

Roles: Olympiad coordinator; Trainer; Association executive-committee member

Mathematical domains: Inequalities; Number theory; AM–GM; Cauchy–Schwarz inequality

Evidence trail

- L1 • Reported gaining resources and knowledge from Academy experts.
- L3 • Defined the participant role as grasping, practising, and re-explaining rather than merely receiving content.
- L3 • Reported increased capability and confidence in online training and later serving as a francophone trainer for African students.
- L5 • Students were reported as more comfortable with inequalities and number theory and able to use AM–GM and Cauchy–Schwarz appropriately.

Artifact status

No training materials or student work were supplied.

INTERPRETIVE BOUNDARY

The account supports refinement within an established leadership trajectory, not the creation of that trajectory by MOTA alone.

6 • PARTICIPANT EVIDENCE ATLAS | P10 • ANONYMIZED CASE

Reusing Academy exercises across in-person and remote training

POSITION	L3 • Self-reported enacted transfer	MOTA LINK	Strong
PRIMARY	Community multiplication	LEVELS	L3

The participant reported several face-to-face and online training circles and camps, serving as a trainer and using exercises from Academy sessions.

Professional context

Roles: Olympiad trainer

Mathematical domains: Academy exercises; Problem solving

Evidence trail

- L3 • Used several Academy exercises in face-to-face and remote training activities.

Artifact status

The reused exercises and training materials were not supplied.

INTERPRETIVE BOUNDARY

The response supports enacted use but does not identify a specific learner-facing result.

6 • PARTICIPANT EVIDENCE ATLAS | P11 • ANONYMIZED CASE

Establishing the first Mathematics Olympiad in Anbar

POSITION	L3 • Self-reported enacted transfer	MOTA LINK	Weak
PRIMARY	System building	LEVELS	L3

The participant coordinated and supervised the first Mathematics Olympiad in Anbar, involving four schools and using simulations modelled on regional and international competitions.

Professional context

Roles: Olympiad coordinator; Competition supervisor; Question-development supervisor

Mathematical domains: Combinatorial games; Invariants; Cauchy–Bunyakovsky–Schwarz inequality

Evidence trail

- L3 • Coordinated the first Mathematics Olympiad in Anbar and supervised question development and four-school delivery.

Artifact status

Competition materials and question sets were described but not supplied.

INTERPRETIVE BOUNDARY

The initiative is specific, but the response does not state that MOTA caused or materially shaped the competition.

6 • PARTICIPANT EVIDENCE ATLAS | P12 • ANONYMIZED CASE

Connecting Academy methods to national-team training and student confidence

POSITION	L5 • Learner-facing signal	MOTA LINK	Strong
PRIMARY	Community multiplication	LEVELS	L3, L4, L5

The participant described three-stage selection, intensive Libyan team training, and use of CBS, Euler, and Fermat. Students were reported as using Fermat enthusiastically for last-digit problems.

Professional context

Roles: National Olympiad committee member; National-team trainer

Mathematical domains: Cauchy–Bunyakovsky–Schwarz inequality; Euler’s theorem; Fermat’s theorem; Number theory

Evidence trail

- L3 • Reported training the team on material studied through the Academy and using CBS in proofs.
- L5 • Students used Fermat’s theorem enthusiastically for last-digit and last-two-digit problems.
- L4 • Reported sending a student competition through Google Classroom.
- L4 • A supplementary message reported an assessment and a best student response after training.
- L5 • Reported a gold medal in an Arab mathematics-games and logic competition.

Artifact status

A Google Classroom competition, an assessment, and a best student response were referenced, but the files were not embedded in the source and were not inspected.

INTERPRETIVE BOUNDARY

The learner behavior is a meaningful self-report. The competition, assessment, and medal references require the absent source materials before functioning as inspected evidence.

6 • PARTICIPANT EVIDENCE ATLAS | P13 • ANONYMIZED CASE

Building a national challenge structure while deepening classroom explanation

POSITION	L5 • Learner-facing signal	MOTA LINK	Mixed
PRIMARY	System building	LEVELS	L3, L5

The participant described a three-person team preparing three-stage Yemeni Olympiad questions across two seasons while participating in MOTA and emphasized deeper concepts and connections through cyclic-quadrilateral teaching.

Professional context

Roles: Scientific-committee chair; Olympiad question designer

Mathematical domains: Geometry; Cyclic quadrilaterals

Evidence trail

- L3 • Helped prepare multi-stage Olympiad questions across two seasons.
- L3 • Deepened conceptual connections and attention to detail through cyclic-quadrilateral teaching.
- L5 • Students showed increased desire to develop in Olympiad mathematics and requested references and training materials.
- L5 • Students achieved local first places; the participant stated that direct coaching was not permitted because of the scientific leadership role.
- L5 • The national initiative was reported as targeting more than 200,000 students.

Artifact status

Olympiad question sets, training references, and reach records were described but not supplied.

INTERPRETIVE BOUNDARY

The reflection distinguishes concurrent Academy participation from direct causation and disclaims direct coaching of the prize-winning students. The reach figure was not independently inspected.

6 • PARTICIPANT EVIDENCE ATLAS | P14 • ANONYMIZED CASE

Teach-back preparation becoming material for student use

POSITION	L5 • Learner-facing signal	MOTA LINK	Strong
PRIMARY	Classroom translation	LEVELS	L3, L5

The participant reported reteaching an Academy session, revising Nim material while preparing a participant presentation, and later giving the revised material to students.

Professional context

Roles: Olympiad trainer

Mathematical domains: Nim; Geometry; Power of a point

Evidence trail

- L3 • Prepared and delivered a participant presentation, then reused the revised content with students.
- L3 • Reported prior involvement training students preparing for the IMO.
- L5 • Students were excited by power of a point, explored with technology, and persisted after hints to find solutions independently.

Artifact status

The participant presentation and revised student material were described but not supplied.

INTERPRETIVE BOUNDARY

The response offers a direct teach-back-to-student-use pathway; student learning itself was not independently assessed.

6 • PARTICIPANT EVIDENCE ATLAS | P15 • ANONYMIZED CASE

Making strategic orientation precede execution and calculation

POSITION	L5 • Learner-facing signal	MOTA LINK	Strong
PRIMARY	Classroom translation	LEVELS	L1, L3, L5

The participant highlighted Cycle I's emphasis on the heuristic dimension of problem solving and reported using that orientation with students before they begin execution and calculation.

Professional context

Roles: National-team deputy leader; International competition team leader; Gifted-student camp trainer

Mathematical domains: Problem-solving heuristics; Strategic problem solving

Evidence trail

- L1 • Identified the heuristic dimension of problem solving—not only mathematical content—as the lasting idea from Cycle I.
- L3 • Applied a deliberate pause before execution and calculation, asking students to decide where to look.
- L5 • Reported that students now pause before calculating and ask, 'Where should I look?'

Artifact status

No lesson material, observation record, or student work was supplied.

INTERPRETIVE BOUNDARY

The reflection supports a specific change in students' problem-solving stance as observed by the participant; no independent observation or student work was supplied.

6 • PARTICIPANT EVIDENCE ATLAS | P16 • ANONYMIZED CASE

Simplifying inequality ideas within Libyan team training

POSITION	L5 • Learner-facing signal	MOTA LINK	Mixed
PRIMARY	Community multiplication	LEVELS	L1, L3, L5

The participant described training Libyan students for mathematics and logic competitions, simplifying inequalities through foundational laws, and reporting strong competition performance.

Professional context

Roles: Libyan team trainer

Mathematical domains: Geometry; Inequalities; Number theory; Arithmetic and geometric means

Evidence trail

- L1 • Identified simplification of inequalities and early introduction of arithmetic-mean and geometric-mean laws as useful learning.
- L3 • Reported applying these ideas while training a Libyan team for mathematics and logic competitions.
- L5 • Reported a gold medal in the Arab competition and a high result in the international competition.

Artifact status

Competition results and supporting records were not supplied for inspection.

INTERPRETIVE BOUNDARY

The result is participant-reported. The source does not independently verify the result or isolate MOTA’s contribution to the team’s performance.

7 • EVIDENCE STATEMENT

What the current evidence establishes

DELIVERY	17 participant-facing sessions • 51 scheduled hours	ESTABLISHED
ENGAGEMENT	354 recorded attendances • 681 participant-hours	ESTABLISHED
TEACHER AGENCY	Teach-backs plus reports of explaining, adapting, designing, training, and coordinating	SUPPORTED
TRANSFER	15 of 16 cases contain specific enacted classroom, training, community, or system use	SELF-REPORTED EVIDENCE
DOWNSTREAM OUTCOMES	12 of 16 cases contain learner-facing signals	NOT ESTABLISHED

Delivery

Established across four cycles, 13 coach-led sessions, four participant presentation sessions, and 51 scheduled live hours.

Engagement

Established through 354 recorded attendances and 681 participant-hours.

Teacher agency

Supported by the built-in teach-back structure and by accounts of re-explaining, adapting, designing, training, coordinating, and creating.

Transfer

Supported within the reflection set by 15 of 16 cases describing specific enacted use.

Learner-facing evidence

Present in 12 cases as participant-reported behavior, work, participation, or results.

Downstream outcomes

Not established because the reflection set is self-selected, most learner evidence is reported by participants, and the cited artifacts and independent comparison data were not available for inspection.

Evidence boundaries

- The 16 cases describe the participants who supplied reflections; they are not a prevalence estimate for all registered teachers.

- Several participants held established Olympiad leadership roles before joining MOTA. The report distinguishes refinement or extension from the creation of those roles.
- Statements about grades, medals, competition performance, project production, assessments, and large reach figures remain participant reports unless supported by inspected records.
- The analysis records temporal sequence and explicit participant attribution. It does not infer causality from timing alone.
- The absence of a supplied artifact means that documentation was unavailable in the analyzed source; it does not mean that the described activity did not occur.

EVIDENCE POSITION

MOTA has established sustained delivery and engagement. Its reflection record supports teacher agency and enacted transfer and contains early learner-facing signals. Independent, representative, and causal downstream outcomes are not established by the current dataset.

CONCLUSION

The legacy is an architecture for mathematical circulation

The value of MOTA is not captured by session counts alone. The Academy creates a recurring place where advanced mathematics, pedagogical translation, and professional agency meet. Teachers encounter expert mathematical practice, return to explain it, and report carrying it into settings that range from a single classroom to coach development, national-team preparation, question design, camps, competitions, and emerging institutional structures.

This sustained architecture distinguishes the model from episodic professional development. The unit of design is the cycle: study, return, explain, adapt, and use. The evidence shows that this movement is visible in the reflection set. Fifteen of 16 cases describe enacted use, and 12 include learner-facing signals. Those signals concern how learners approach mathematics—where they look, how long they persist, which strategies they try, whether they explain, and what they produce—as well as a smaller number of participant-reported results.

The evidence boundary is part of the conclusion. The reflection record does not establish representative student outcomes or isolate the Academy's causal effect. It does establish that the programme has moved beyond delivery into a documented architecture of teacher agency and participant-reported transfer. The next evidentiary step would be independent documentation of the learner-facing work already described: supplied materials, student solutions, assessments, observations, participation records, and verified results.

Olympiad mathematics becomes shared culture when teachers are given both deeper mathematics and the agency to translate it.

End of report

APPENDIX A • METHOD

Coding and data dictionary

The analysis is a structured qualitative evidence synthesis of participant reflections, read alongside administrative delivery and engagement records. It is not an experimental or quasi-experimental evaluation.

Term	Definition
Submission	One form response. The source contains 17 submissions.
Case	One unique participant after duplicate consolidation. The analysis contains 16 cases.
Meaning unit	A statement or passage that supports one interpretable evidence claim.
Highest position	The farthest evidence level supported anywhere in a case; used for distance on the map.
Level presence	Every level supported within a case. Multiple levels may be present.
Primary constellation	The principal professional-action category used to colour the map.
Any membership	All professional-action categories supported by a case; non-exclusive.
MOTA link	The explicitness of the connection between reported action and Academy content or structure.
Artifact status	Whether a material was described, referenced, supplied, and/or inspected.
Boundary	The qualification needed to keep interpretation within the supplied evidence.

Case-count checks

- 17 submissions were consolidated into 16 unique participant cases.
- All 16 cases appear in the topography and the atlas.
- Highest positions sum to 16: L0 = 1, L3 = 3, L5 = 12.
- Level-presence counts are non-exclusive: L0 = 2, L1 = 5, L2 = 1, L3 = 15, L4 = 1, L5 = 12.
- Professional-action memberships are non-exclusive: classroom translation = 12, community multiplication = 11, system building = 6, emerging evidence = 1.

Source statement

Programme figures derive from the AY 2025-26 schedule and attendance analysis. Qualitative claims derive from the teacher-reflection source, with duplicate responses consolidated and supplementary participant messages attached only where their case connection was identifiable. The report does not add names, institutions, results, documents, or causal claims that are absent from the supplied record.

PRIVACY

P01-P16 are analytic labels. The report omits participant names and email addresses.